Predicting Interstitial Electron Localization and Void Connectivity for Electride Screening

A void-centered machine-learning and geometric framework for characterizing electrides

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LOGO

QR CODE

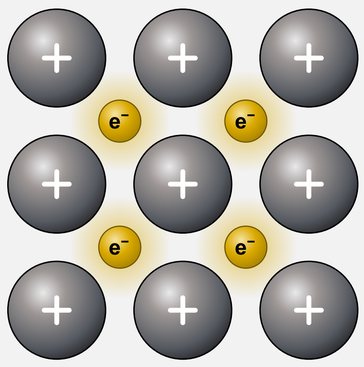
BACKGROUND

What is an electride?

A crystal where electrons act as anions. Extra electrons sit in the voids between atoms.

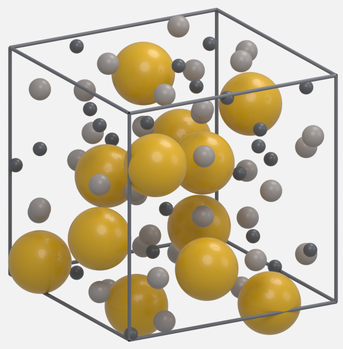
ELF marks them. The electron localization function runs from 0 to 1. A high value in a void means a trapped electron.

Finding them is slow. Each crystal needs its own DFT calculation.



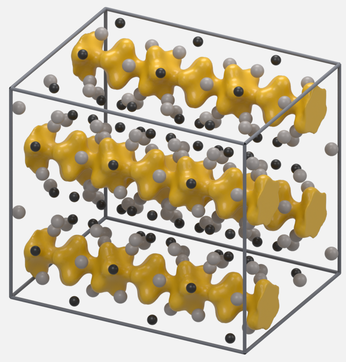
Connectivity defines dimensionality in electrides

0D Isolated pockets



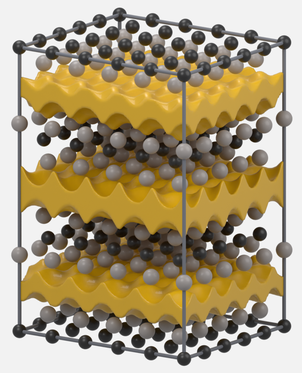
Mayenite-type $Ca_6Al_7O_{16}$: separate cages

1D Channels



Y_5Si_3: channels along one direction

2D Sheets



Ca_2N: layers between the atoms

Gold: void regions **Gray and black:** atoms Images simplified for clarity

GOAL screen crystals for electrides from structure alone, without DFT on every candidate. We built two void-centered tools and tested each one separately.

TOOL 1 · PREDICT ELF

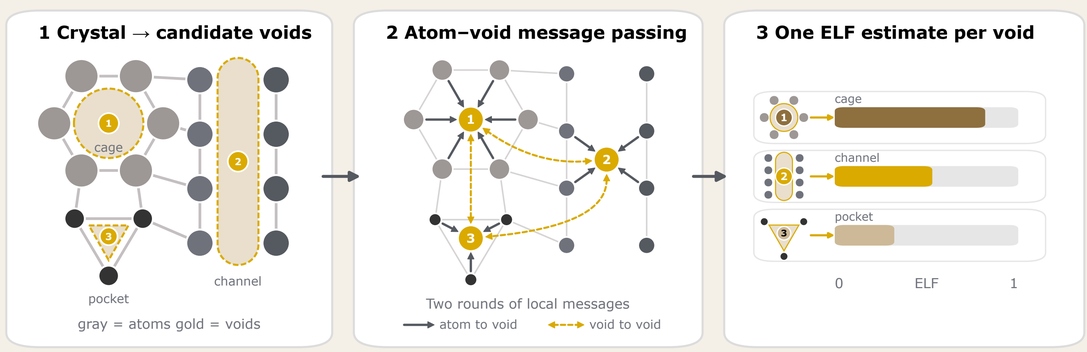
A graph neural network predicts ELF at individual voids from the surrounding atoms

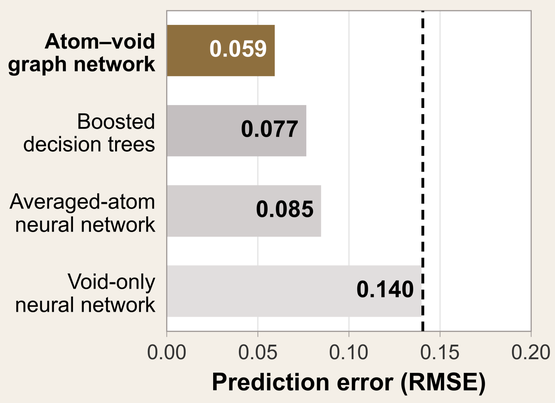
0.82
R^2 on unseen families

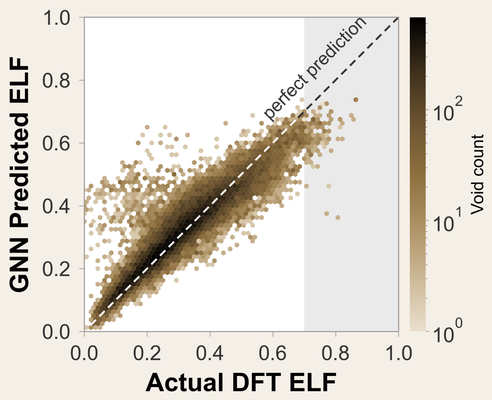
0.059
RMSE on the 0–1 ELF scale

88,680
test voids, 5,000 crystals

ELF at each void from nearby atoms







- **Atoms carry the signal:** a void alone predicts almost nothing (RMSE 0.140).
- **Limits:** trained on ELFNet (compressed hydrides), so not yet tested on known electrides. Rare voids with ELF ≥ 0.7 are under-predicted.

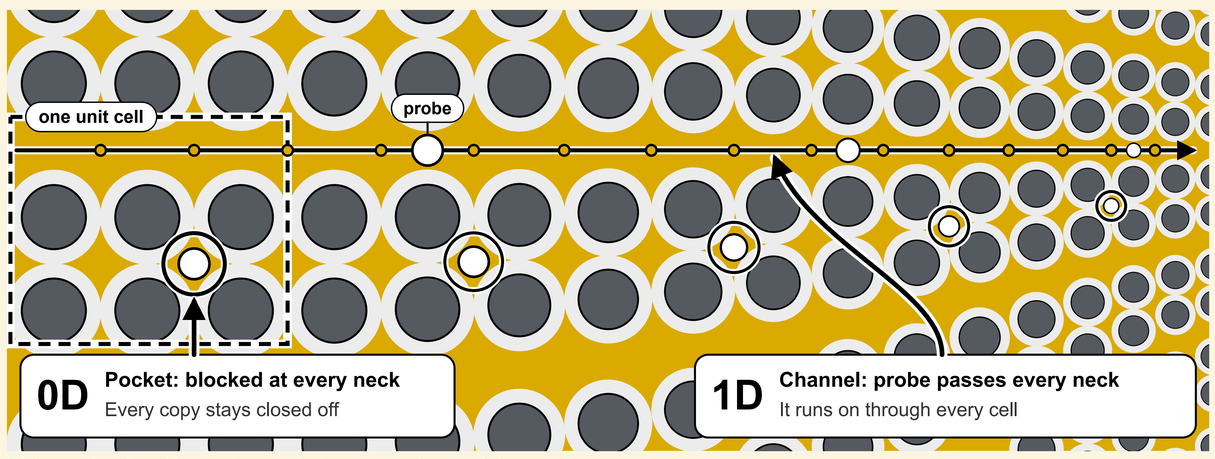
TOOL 2 · MAP HOW VOIDS CONNECT

Geometry alone recovers the dimensionality of known electrides above chance

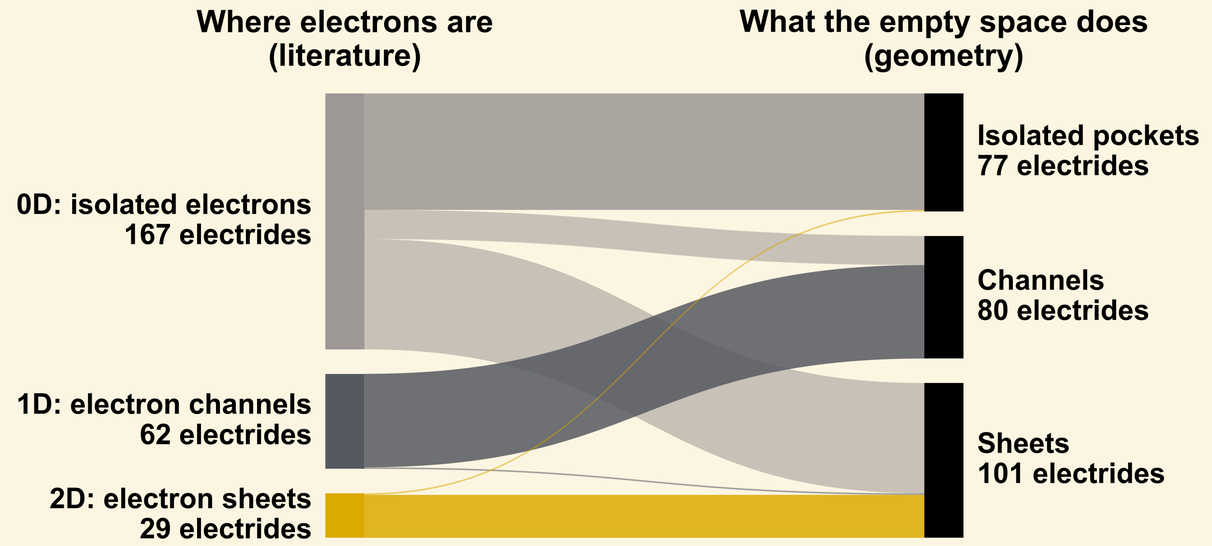
Probe must fit to pass through a neck

Atom + probe radius blocked: probe center cannot enter

Open space where the probe can sit



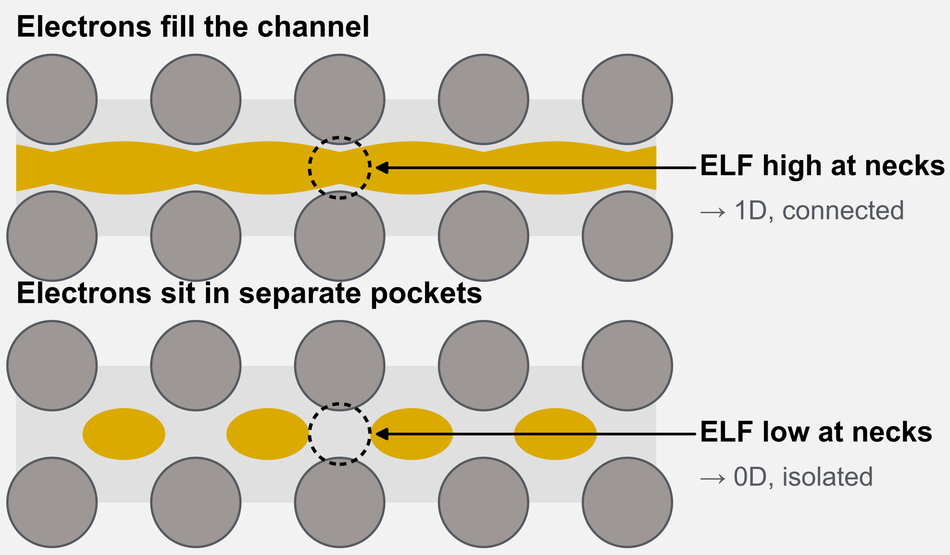
Schematic of an illustrative crystal, not a real material



TOOLS TOGETHER

Open space is not the same as electrons

Same open channel, different electrons: only ELF at the necks tells 1D from 0D.



TAKE-HOME

Predicting ELF at voids and mapping void connectivity build a screen that can separate structural from electronic dimensionality.

A foundation, not yet a validated electride classifier.

What this work adds

- A prediction target at specific crystal voids
- An atom–void graph that keeps chemical and geometric context
- An exact connectivity screen for void networks
- A path to combining geometry with electronic evidence

NEXT STEPS

- Run DFT on known electrides to test the GNN
- Predict ELF at void necks
- Build a crystal-level electride score

REFERENCES Becke & Edgecombe (1990) · Burton et al. (2018) · Zhu et al. (2019) · Zeo++ (2012)